ADS7057 test board (low cost)

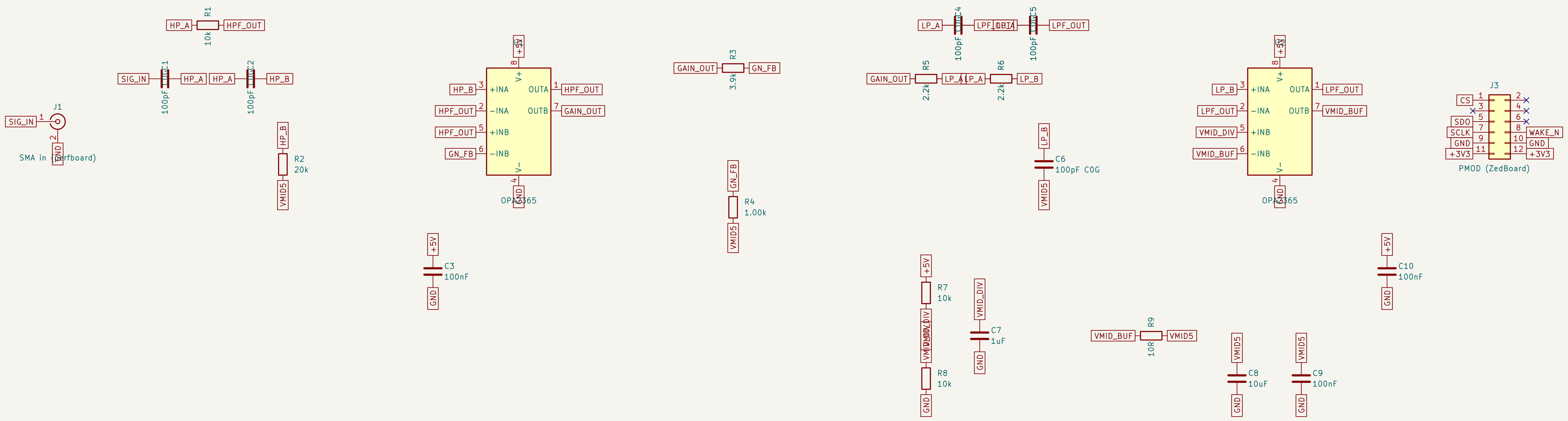
Perfboard charge amp (SMA) -> 2nd-order HPF 113 kHz -> gain x4.9 -> 2nd-order LPF 512 kHz -> THS4531 -> 10R/400pF -> ADS7057 -> ZedBoard PMOD.

AFE v0.1 topology rebuilt around JLC Basic parts (100 pF C0G, E12 resistors, AMS1117, LM393). Power: 9 V -> AMS1117-5.0 -> AMS1117-3.3 (AVDD). DVDD = PMOD 3.3 V.

Input, high-pass filter (Sallen-Key, 100 pF / 10k / 20k: f0 = 113 kHz, Q = 0.71), gain 1 + 3.9k/1k = 4.9

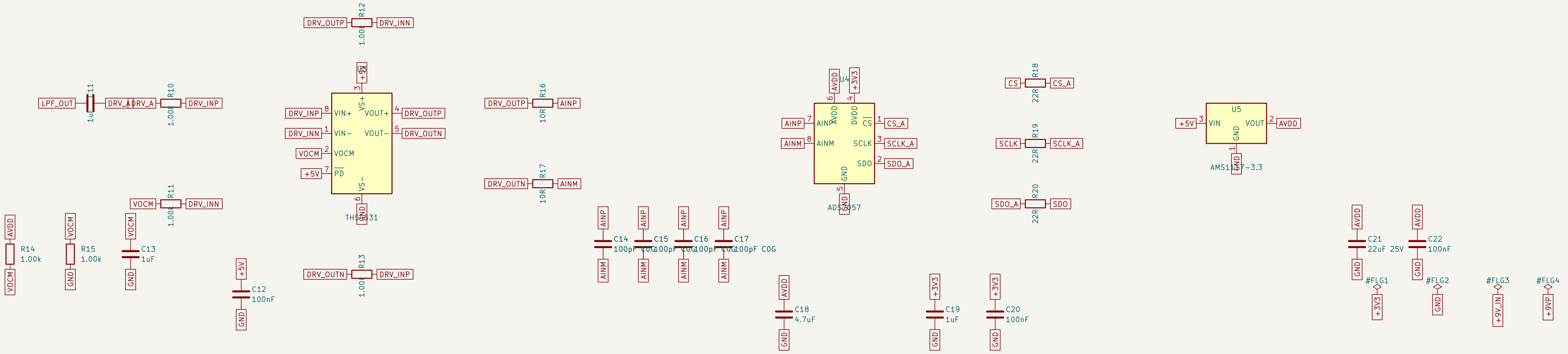
Low-pass filter (2.2k / 2.2k, C1 = 2 x 100 pF, C2 = 100 pF: f0 = 512 kHz, Q = 0.71), VMID5

ZedBoard PMOD JA (3.3 V). Pin 6/12 = 3.3 V from the ZedBoard -> DVDD. Pin 10 = WAKE_N (active low).



Differential driver (gain 1), ADC. Kickback C = 4 x 100 pF C0G = 400 pF (25 x the 16 pF sampling C).

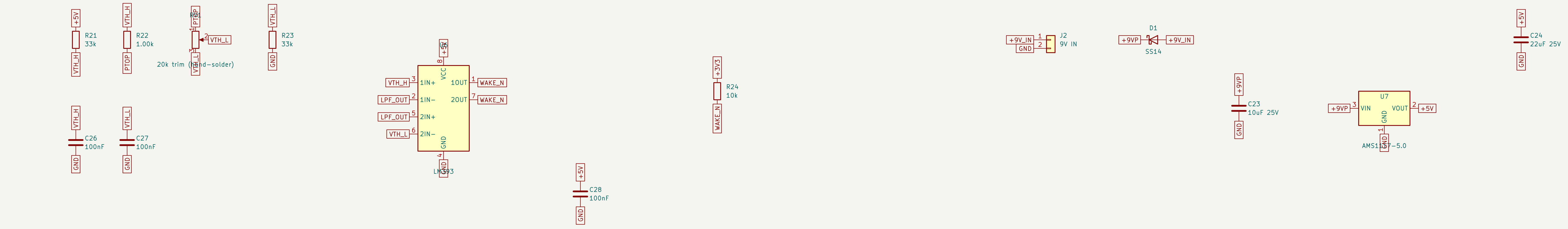
AVDD = AMS1117-3.3 from 5 V (also the ADC reference)



Window comparator / wake-up. Half window = 2.5 V x (1k + Rp)/(67k + Rp) = 37 ... 603 mV with the 20k RV1 (about 7 ... 120 mV at the perfboard output).

LM393 open-collector outputs are wired together: WAKE_N (PMOD pin 10) is pulled up to 3.3 V and goes LOW when the signal leaves the window.

Power: 9 V battery -> SS14 -> AMS1117-5.0 (+5V for op-amps, driver, comparator, AVDD regulator).



Universidad de los Andes – Auselia

Sheet: /
File: ADC_Test.kicad_sch

Title: Auselia ADS7057 test board, low-cost version (JLC Basic parts)

Size: A2 | Date: 2026-10-12 | **Rev: A**
KiCad E.D.A. 10.0.6 | Id: 1/1